

Lab Report-01

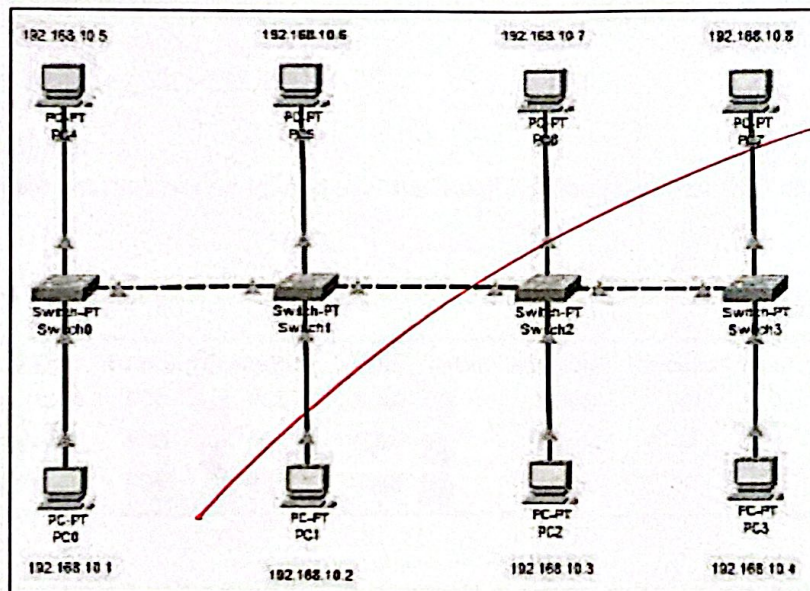
Problem: Bus Topology

Description:

Bus topology is a network setup where all devices are connected to a single central cable (the bus). Data travels in both directions along the bus, but only one device can transmit at a time to avoid collisions. Terminators at both ends of the cable prevent signal reflection. It's inexpensive and easy to install for small networks but hard to troubleshoot and less efficient with high traffic or many devices.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1. First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image.
2. Use Automatic connecting Cable to connect devices with others or individually use the Copper Straight-through Cable (Ethernet) to connect the devices and Use Copper-Cross Cable to connect the switches.
3. Configure the PCs (Hosts) with IPv4 Addresses and Subnet Mask according to the IP addressing Table given below.

4. Assign IP Address:

PC0-192.168.10.1

PC1-192.168.10.2

PC2-192.168.10.3

PC3-192.168.10.4

PC4-192.168.10.5

PC5-192.168.10.6

PC6-192.168.10.7

PC7-192.168.10.8

Set the Subnet Mask (e.g. 255.255.255.0)

5. Test Connectivity:

Open a command prompt on one PC and use the "ping" command to test the connectivity to another PC.

6. Test File Sharing:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC5	ICMP		0.000	N	0	(edit)	
	Successful	PC0	PC7	ICMP		0.000	N	1	(edit)	
	Successful	PC5	PC3	ICMP		1.988	N	2	(edit)	

Conclusion:

Bus topology is a straightforward and economical networking structure where all devices are connected to a single central cable. It is simple and cost-effective, ideal for small networks. It suffers from performance issues and is hard to troubleshoot as more devices are added. A single cable failure can bring down the entire network.

Lab Report-02

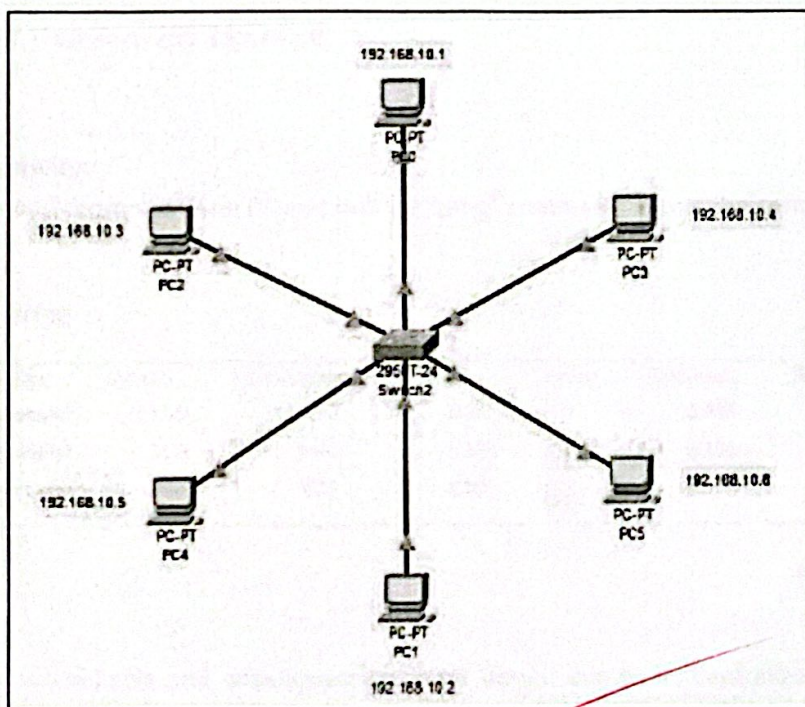
Problem: Star Topology

Description:

Star topology is a network configuration where all devices are individually connected to a central device (such as a switch or hub). The central device manages and controls all data traffic. If one device falls, the rest remain unaffected, but if the central device fails, the entire network goes down. It offers better performance, easier troubleshooting, and is more scalable than bus topology but requires more cable and is costlier.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1. First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Hub) and Connections (Cables) as shown above the image.
2. Use Automatic connecting Cable to connect devices with others or individually use the Copper Straight-through Cable (Ethernet) to connect the devices and Use Copper-Cross Cable to connect the switches.

3. Configure the PCs (Hosts) with IPv4 Addresses and Subnet Mask according to the IP addressing Table given below.

4. Assign IP Address:

PC0-192.168.10.1

PC1-192.168.10.2

PC2-192.168.10.3

PC3-192.168.10.4

PC4-192.168.10.5

PC5-192.168.10.6

Set the Subnet Mask (e.g. 255.255.255.0)

5. Test Connectivity:

Open a command prompt on one PC and use the "ping" command to test the connectivity to another PC.

6. Test File Sharing:

File	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	PC3	ICMP		0.000	N	0
	Successful	PC0	PC5	ICMP		0.106	N	1
	Successful	PC0	PC2	ICMP		2.104	N	2

Conclusion:

Star topology is a reliable and widely used network design due to its centralized structure. It offers easy troubleshooting, high performance, and minimal impact from individual device failures. The central hub or switch efficiently manages data flow between devices. However, the entire network depends on the central device-if it fails, the whole system is affected.

Lab Report-03

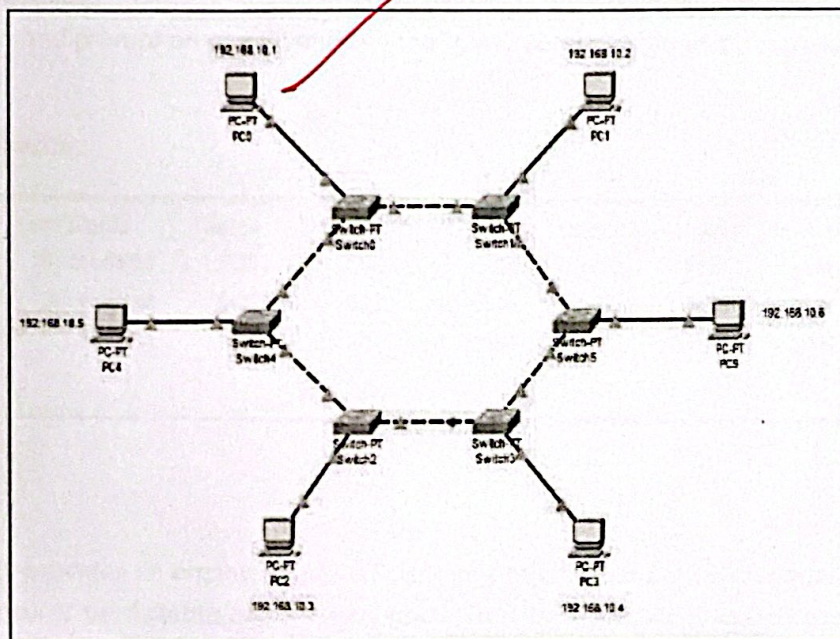
Problem: Ring Topology

Description:

Ring topology is a network configuration where each device is connected to exactly two other devices, forming a closed loop or ring. Data travels in one direction, passing through each device until it reaches its destination. Each device acts as a repeater, boosting the signal to ensure it reaches the next node. It offers orderly data transfer and predictable performance but is vulnerable to failure-if one device or connection breaks, the whole network can be disrupted.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1. First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image.
2. Use Automatic connecting Cable to connect devices with others or individually use the Copper Straight-through Cable (Ethernet) to connect the devices and Use Copper-Cross Cable to connect the switches.
3. Configure the PCs (Hosts) with IPv4 Addresses and Subnet Mask according to the IP addressing Table given below.

4. Assign IP Address:

PC0-192.168.10.1

PC1-192.168.10.2

PC2-192.168.10.3

PC3-192.168.10.4

PC4-192.168.10.5

PC5-192.168.10.6

Set the Subnet Mask (e.g. 255.255.255.0)

5. Test Connectivity:

Open a command prompt on one PC and use the "ping" command to test the connectivity to another PC.

6. Test File Sharing:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC5	PC2	ICMP		0.000	N	0
	Successful	PC5	PC3	ICMP		0.003	N	1
	Successful	PC0	PC4	ICMP		0.005	N	2

Conclusion:

Ring topology provides an organized and efficient way to transmit data with equal access for all devices. It ensures predictable performance under light to moderate loads. However, the main drawback is its dependency on each connection-if a single device or link fails, the entire network can be affected. Despite this, using a dual ring can improve reliability and fault tolerance.

Lab Report-04

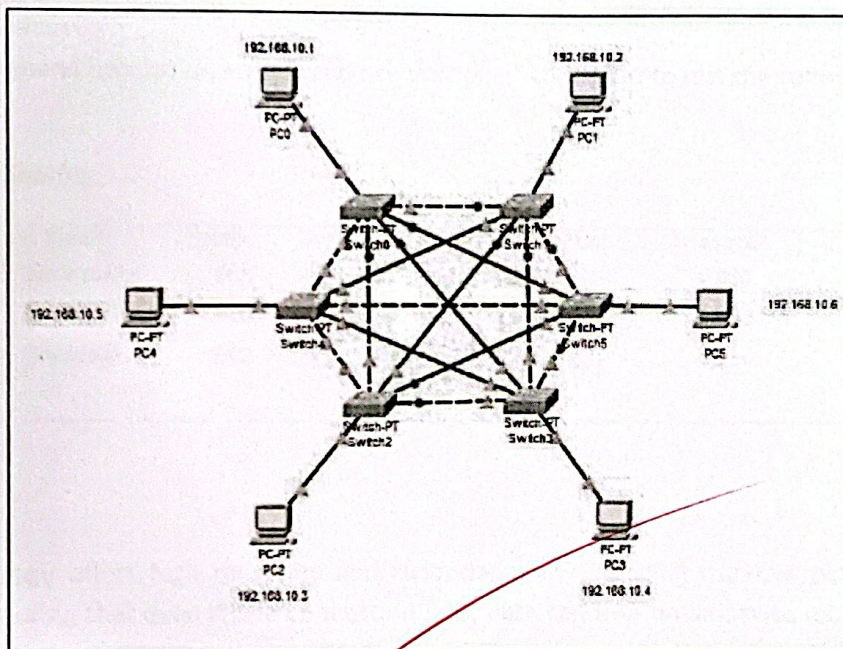
Problem: Mesh Topology

Description:

Mesh topology connects each device to every other device in the network, ensuring multiple paths for data transmission. This provides high reliability and fault tolerance, as data can still flow even if one link fails. It is ideal for critical systems needing constant uptime. However, it is costly and complex due to extensive cabling and configuration.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1. First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image.
2. Use Automatic connecting Cable to connect devices with others or individually use the Copper Straight-through Cable (Ethernet) to connect the devices and Use Copper-Cross Cable to connect the switches.
3. Configure the PCs (Hosts) with IPv4 Addresses and Subnet Mask according to the IP addressing Table given below.

4. Assign IP Address:

PC0-192.168.10.1

PC1-192.168.10.2

PC2-192.168.10.3

PC3-192.168.10.4

PC4-192.168.10.5

PC5-192.168.10.6

Set the Subnet Mask (e.g. 255.255.255.0)

5. Test Connectivity:

Open a command prompt on one PC and use the "ping" command to test the connectivity to another PC.

6. Test File Sharing:

ire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC5	PC1	ICMP		0.000	N	0
	Successful	PC0	PC4	ICMP		0.002	N	1
	Successful	PC2	PC5	ICMP		0.004	N	2

Conclusion:

Mesh topology offers high reliability and redundancy by providing multiple paths between devices, ensuring that even if one connection fails, data can take an alternate route. It is ideal for critical systems where continuous network availability is essential. However, it is complex to install, requires more cabling, and is costly to maintain. Despite the expense, its robustness makes it suitable for large, mission-critical networks.

Lab Report-05

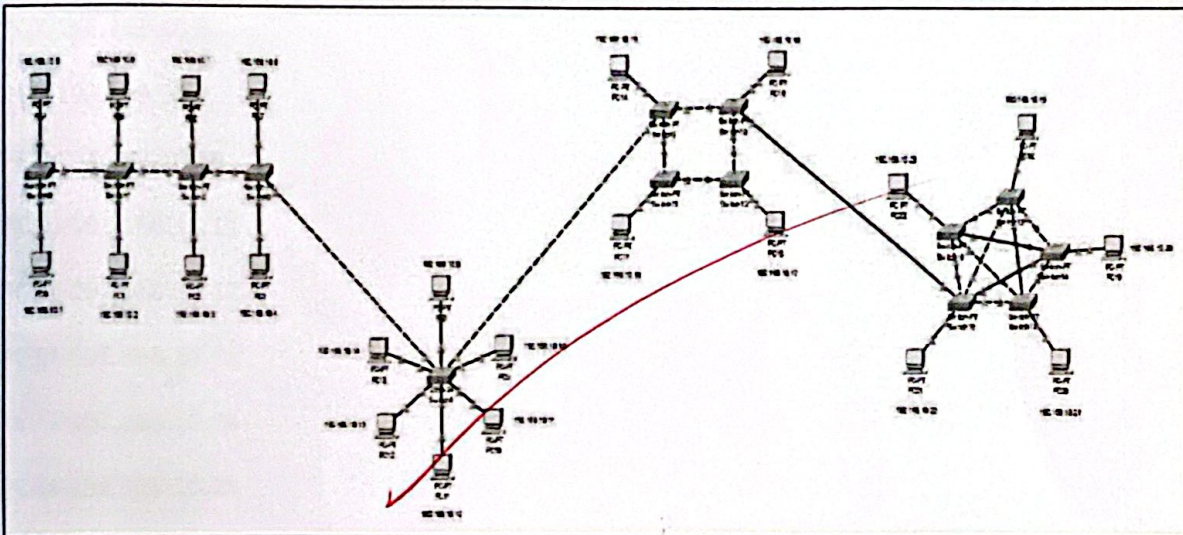
Problem: Hybrid Topology

Description:

Hybrid topology combines two or more basic topologies (like star, bus, or ring) to create a flexible and scalable network, it benefits from the strengths of each topology used, offering better performance and reliability. This makes it ideal for large, complex systems. However, it can be expensive and difficult to manage

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1. First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image.
2. Use Automatic connecting Cable to connect devices with others or individually use the Copper Straight-through Cable (Ethernet) to connect the devices and Use Copper-Cross Cable to connect the switches.
3. Configure the PCs (Hosts) with IPv4 Addresses and Subnet Mask according to the IP addressing Table given below.

4. Assign IP Address:

PC0-192.168.10.1

PC1-192.168.10.2

PC2-192.168.10.3

PC3-192.168.10.4

PC4-192.168.10.5

PC5-192.168.10.6

PC6-192.168.10.7

PC7-192.168.10.8

PC8-192.168.10.9

PC9-192.168.10.10

PC10-192.168.10.11

PC11-192.168.10.12

PC12-192.168.10.13

PC13-192.168.10.14

PC14-192.168.10.15

PC15-192.168.10.16

PC16-192.168.10.17

PC17-192.168.10.18

PC18-192.168.10.19

PC19-192.168.10.20

PC20-192.168.10.21

PC21-192.168.10.22

PC22-192.168.10.23

Set the Subnet Mask (e.g. 255.255.255.0)

5. Test Connectivity:

Open a command prompt on one PC and use the "ping" command to test the connectivity to another PC.

6. Test File Sharing:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
●	Successful	PC0	PC18	ICMP	■	0.000	N	0
●	Successful	PC11	PC16	ICMP	■	0.005	N	1
●	Successful	PC12	PC3	ICMP	■	0.008	N	2

Conclusion:

Hybrid topology is a highly flexible and scalable network design that combines the strengths of multiple basic topologies like star, bus, and ring. It offers enhanced performance, reliability, and adaptability, making it ideal for large and complex network environments. Although it can be costly and complex to design and manage, its ability to meet diverse networking needs makes it a preferred choice for modern organizations.

Lab Report-06

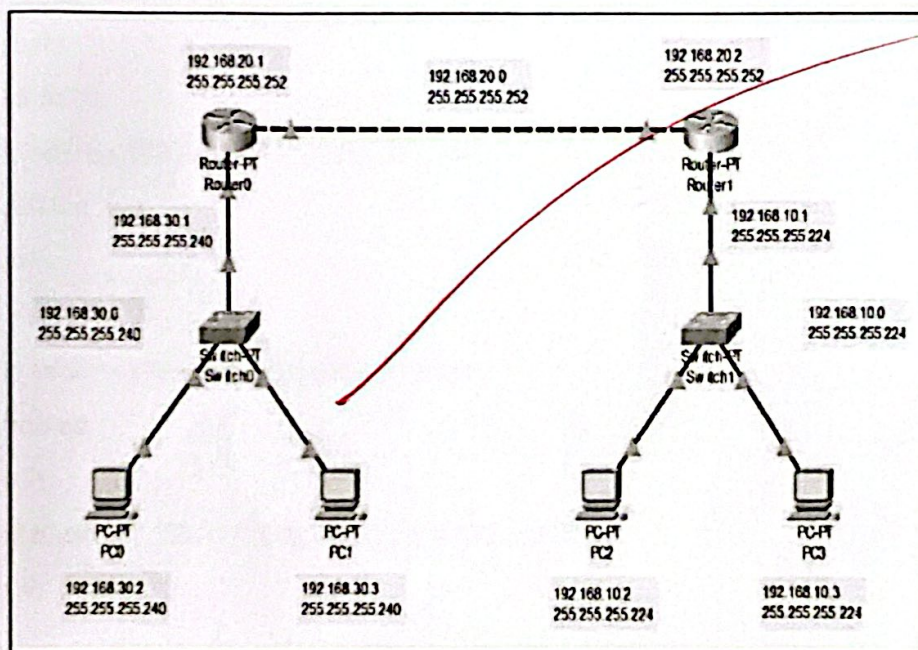
Problem: Static Routing

Description:

Static routing is a method of routing where routes are manually configured and maintained by a network administrator. Each route must be explicitly defined, specifying the destination network and the next-hop address. It is simple and secure, making it suitable for small or stable networks. However, it lacks adaptability—changes in the network require manual updates, which can be time-consuming and error-prone in larger or dynamic environments.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1. First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches and Hubs) and Connections (Cables) as shown above the image.
2. Use Automatic connecting Cable to connect devices with others or individually use the Copper Straight-through Cable (Ethernet) to connect the devices and Use Copper-Cross Cable to connect the switches.

3. Configure the PCs (Hosts) with IPv4 Addresses and Subnet Mask according to the IP addressing Table given below.

4. Assign IP Address:

PC0-192.168.30.2 (Subnet Mask 255.255.255.240)

PC1-192.168.30.3 (Subnet Mask 255.255.255.240)

PC2-192.168.10.2 (Subnet Mask 255.255.255.224)

PC3-192.168.10.3 (Subnet Mask 255.255.255.224)

6. Configuration Setting:

Router-0:

- en
- conf t
- int fa0/0
- ip address 192.168.30.1 255.255.255.240
- no shut
- exit
- int fa1/0
- ip address 192.168.20.1 255.255.255.252
- no shut
- exit
- ip route 192.168.10.0 255.255.255.224 192.168.20.2
- exit

Router-1:

- en
- conf t
- int fa0/0
- ip address 192.168.10.1 255.255.255.224
- no shut
- exit
- int fa1/0

- ip address 192.168.20.2 255.255.255.252
- no shut
- exit
- ip route 192.168.30.0 255.255.255.240 192.168.20.1
- exit

6. Test Connectivity:

Open a command prompt on one PC and use the "ping" command to test the connectivity to another PC.

7. Test File Sharing:

File	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	PC3	ICMP		0.000	N	0
	Successful	PC2	PC1	ICMP		1.998	N	1
	Successful	PC0	PC2	ICMP		4.000	N	2

Conclusion:

Static routing is reliable, secure, and easy to implement in small or stable networks. It offers full control over routing paths with minimal resource usage. However, it lacks flexibility and requires manual changes for any network updates. This limits its efficiency in large or frequently changing networks.

Lab Report-07

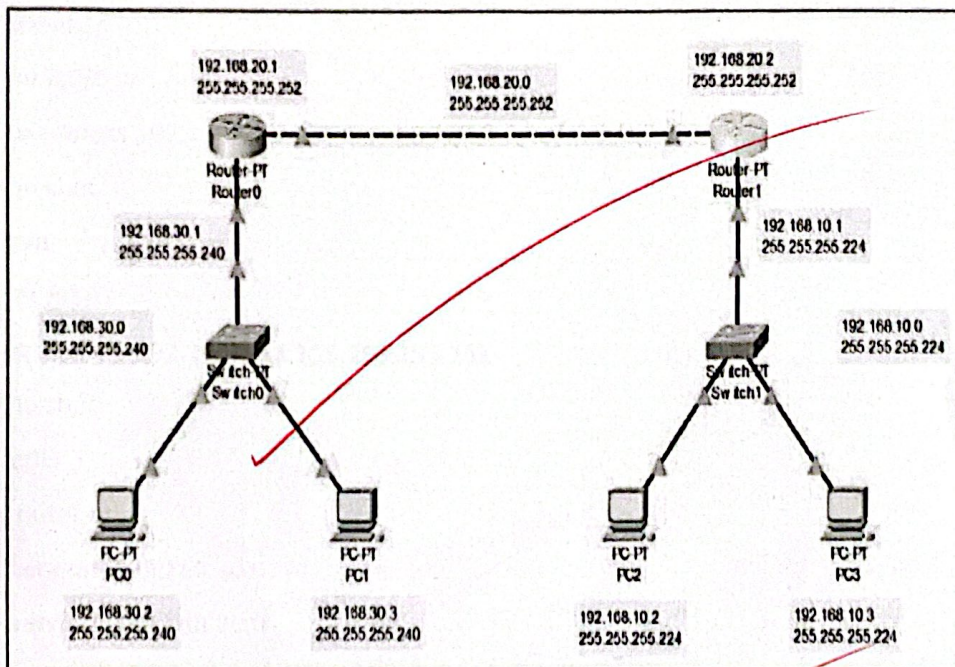
Problem: Dynamic Routing

Description:

Dynamic routing is a method where routers automatically learn and update routes using routing protocols like RIP, OSPF, or EIGRP. It allows the network to adjust to changes, such as link failures or new routes, without manual intervention. This makes it highly suitable for large and complex networks. However, it consumes more bandwidth and processing power than static routing.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1. First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches and Hubs) and Connections (Cables) as shown above the image.
2. Use Automatic connecting Cable to connect devices with others or individually use the Copper Straight-through Cable (Ethernet) to connect the devices and Use Copper-Cross Cable to connect the switches.
3. Configure the PCs (Hosts) with IPv4 Addresses and Subnet Mask according to the IP addressing Table given below.

4. Assign IP Address:

PC0-192.168.30.2 (Subnet Mask 255.255.255.240)

PC1-192.168.30.3 (Subnet Mask 255.255.255.240)

PC2-192.168.10.2 (Subnet Mask 255.255.255.224)

PC3-192.168.10.3 (Subnet Mask 255.255.255.224)

6 (i). Configuration Setting [RIP Version 1]:

Router-0:

- en
- conf t
- int fa0/0
- ip address 192.168.30.1 255.255.255.240
- no shut
- exit
- int fa1/0
- ip address 192.168.20.1 255.255.255.252
- no shut
- exit
- router rip
- network 192.168.30.0
- network 192.168.20.0
- exit

Router-1:

- en
- conf t
- int fa0/0
- ip address 192.168.10.1 255.255.255.224
- no shut
- exit
- int fa1/0

- ip address 192.168.20.2 255.255.255.252
- no shut
- exit
- router rip
- network 192.168.10.0
- network 192.168.20.0
- exit

(ii). Configuration Setting [RIP Version 2]:

Router-0:

- conf t
- int fa0/0
- ip address 192.168.30.1 255.255.255.240
- no shut
- exit
- int fa1/0
- ip address 192.168.20.1 255.255.255.252
- no shut
- exit
- router rip
- version 2
- network 192.168.30.0
- network 192.168.20.0
- no auto summary
- exit

Router-1:

- en
- conf t
- int fa0/0

- ip address 192.168.20.2 255.255.255.252
- no shut
- exit
- int fa1/0
- ip address 192.168.10.1 255.255.255.224
- no shut
- exit
- router rip
- version 2
- network 192.168.10.0
- network 192.168.20.0
- no auto summary
- exit

6. Test Connectivity:

Open a command prompt on one PC and use the "ping" command to test the connectivity to another PC.

7. Test File Sharing:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	PC3	ICMP		0.000	N	0
	Successful	PC2	PC1	ICMP		0.000	N	1
	Successful	PC1	PC3	ICMP		0.010	N	2

Conclusion:

Dynamic routing automatically updates routing tables based on network changes, making it ideal for large and frequently changing networks. It reduces manual effort and adapts quickly to link failures or topology changes. Though it uses more bandwidth and resources, its flexibility and efficiency outweigh the cost. Thus, it is widely used in modern, scalable network environments.

Lab Report-08

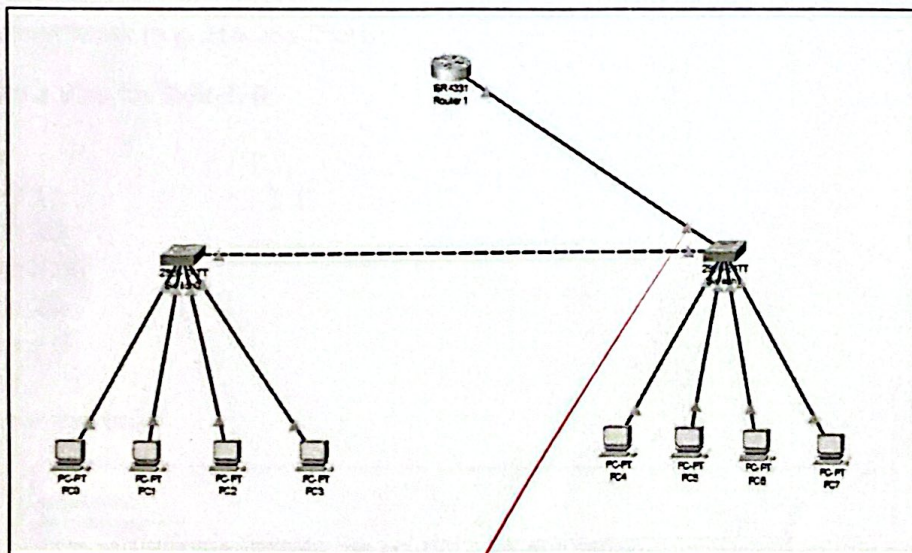
Problem: Vlan

Description:

A VLAN is a logical subdivision of a physical network that groups devices based on function, department, or application, regardless of their physical location. It allows devices to communicate as if they were on the same LAN, even when connected to different switches. VLANs enhance security by isolating traffic and reduce broadcast domains, improving network performance. They are widely used in enterprise networks for better traffic management and scalability.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1. First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches and Hubs) and Connections (Cables) as shown above the image.
2. Use Automatic connecting Cable to connect devices with others or individually use the Copper Straight-through Cable (Ethernet) to connect the devices and Use Copper-Cross Cable to connect the switches.
3. Configure the PCs (Hosts) with IPv4 Addresses and Subnet Mask according to the Ip addressing Table given below.

4. Assign IP Address:

PC0-192.168.10.1

PC1-192.168.10.2

PC2-192.168.20.1

PC3-192.168.20.2

PC4-192.168.10.3

PC5-192.168.10.4

PC6-192.168.20.3

PC7-192.168.20.4

Set the Subnet Mask (e.g. 255.255.255.0)

5 (i). Create a Vlan for Switch-0:

- en
- conf t
- vlan 10
- name HR
- vlan 20
- name IT
- end
- show vlan brief

```
Switch>en
Switch>conf t
Enter configuration commands, one per line. End with CTRL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10 HR	active	
20 IT	active	
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```


5 (ii). Assign Port:

- conf t
- int range fa0/1-2
- switchport access vlan 10
- int range fa0/3-4
- switchport access vlan 20
- end
- show vlan brief

```
Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/1-2
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#int range fa0/3-4
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Switch#show vlan brief

VLAN Name	Status	Ports
1 default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10 HR	active	Fa0/1, Fa0/2
20 IT	active	Fa0/3, Fa0/4
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Switch#

- show interface switchport

```
Switch#show interface switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
```

5 (iii). Convert Dynamic to Static:

- conf t
- int range fa0/1-4
- switchport mode access
- end
- show interface switchport


```

Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/1-4
Switch(config-if-range)#
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off

```

5 (iv). Trunking:

- en
- conf t
- int range fa0/5
- switchport mode trunk
- end
- show interface trunk

```

Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/5
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/5	on	802.1q	trunking	1

```

Port      Vlans allowed on trunk
Fa0/5     1-1005

Port      Vlans allowed and active in management domain
Fa0/5     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/5     1,10,20

Switch#

```


5 (i). Create a Vlan for Switch-1:

- en
- conf t
- vlan 10
- name HR
- vlan 20
- name IT
- end
- show vlan brief

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	HR	active	
20	IT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
Switch#
```

6 (ii). Assign Port:

- conf t
- int range fa0/1-2
- switchport access vlan 10
- int range fa0/3-4
- switchport access vlan 20
- end
- show vlan brief


```

Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/1-2
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#int range fa0/3-4
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

```

```
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10 HR	active	Fa0/1, Fa0/2
20 IT	active	Fa0/3, Fa0/4
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

→ show interface switchport

```

Switch#show interface switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native

```

6 (iii). Convert Dynamic to Static:

- conf t
- int range fa0/1-4
- switchport mode access
- end
- show interface switchport

```

Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/1-4
Switch(config-if-range)#
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off

```


6 (iv). Trunking:

- en
- conf t
- int range fa0/5
- switchport mode trunk
- end
- show interface trunk

```
Switch#en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range fa0/5
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/5     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/5     1-1005

Port      Vlans allowed and active in management domain
Fa0/5     1,10,20

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/5     1,10,20

Switch#
```

7. Router Configuration Setting:

- en
- conf t
- int gig0/0
- no shut
- int gig0/0.1
- encapsulation dot1q 10
- ip add 192.168.10.5 255.255.255.0
- exit
- int gig0/0.2
- encapsulation dot1q 20
- ip add 192.168.20.5 225.225.225.0
- end

→ show ip int brief

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gig0/0/0
Router(config-if)#no shut

Router(config-if)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#int gig0/0.1
%Invalid interface type and number
Router(config)#int gig0/0.1
Router(config-subif)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0/0.1, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.1, changed state to up

Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip add 192.168.10.5 255.255.255.0
Router(config-subif)#exit
Router(config)#int gig0/0.2
Router(config-subif)#
%LINK-S-CHANGED: Interface GigabitEthernet0/0/0.2, changed state to up

%LINEPROTO-S-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.2, changed state to up

Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip add 192.168.20.5 225.255.255.0
Bad mask 0x22FFFFFF0 for address 192.168.20.5
Router(config-subif)#end
```

8. Test Connectivity:

Open a command prompt on one PC and use the "ping" command to test the connectivity to another PC.

9. Test File Sharing:

File	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC4	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC3	PC6	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC5	PC0	ICMP		0.000	N	2	(edit)	(delete)

Conclusion:

VLAN (Virtual Local Area Network) effectively segments a physical network into multiple logical networks, improving security, performance, and manageability. In a lab environment, VLANs help simulate real-world enterprise setups by allowing multiple departments or groups to operate on separate broadcast domains without additional hardware. They reduce network congestion and provide better control over traffic. Overall, VLANs are a crucial concept in modern networking, making network design more flexible and efficient.